Supplementary Notes S1–S2

Reference-Capacity-Induced Label-Definition Shift in Lithium-Ion Battery SOH Benchmarking and RUL Label Availability: Multi-Dataset Sensitivity, Censoring, and Reproducibility

Supplementary Tables S1–S9 are provided in the workbook `Supplementary_Tables_S1-S9.xlsx`. This document provides Supplementary Notes S1 and S2. All content is taken from frozen project artifacts. No data were reprocessed, no threshold or model was changed, and no endpoint was recalculated.

Supplementary Note S1. Capacity extraction, quality control, and reproducibility

CALCE. Single-cycle discharge capacity was extracted from the CS2 source workbooks without modifying them. Each cycle was checked against charge, constant-voltage, discharge, timing, voltage, and file-boundary evidence. Automated flags were kept alongside manual decisions, and protocol-incomplete cycles were not silently deleted. CALCE was used for method development and QC support, not as an additional reference-capacity sensitivity endpoint.

XJTU and HUST. The analysis used the fixed, validated primary-capacity records and their final inclusion decisions (XJTU Final V2; HUST Final V1). Protocol-incomplete observations and unresolved records were excluded only through the approved QC layers, and their original records and exclusion reasons remain traceable. After inclusion rules were applied, 10,427 primary capacity observations were retained for XJTU and 146,122 for HUST.

Reference values (Table S3). Supplementary Table S3 reports per-cell reference values and source cycles read from the frozen XJTU Final V2 and HUST Final V1 outputs. It supersedes an earlier pre-hotfix table in which the XJTU Batch 2–6 `first5_comparable` source observations had been string-sorted. The per-cell `first5_comparable` values used by the frozen endpoint classifier agree exactly with Table S3 for all 55 XJTU cells, and the primary SOH metrics in Table S4 are identical to those reported in the manuscript. In XJTU, `first5_comparable` has a median of 1.923 Ah (median difference from nominal −0.077 Ah; range −0.195 to −0.038 Ah), and its fifth source observation occurs at raw-cycle indices 6–65 (median 23).

General rules. Capacity trajectories were not smoothed, missing cycles were not interpolated, anomalous capacities were not replaced, and original cycle indices were not compressed or renumbered. Raw source files were not modified.

Software environment. Python 3.12.6, numpy 2.5.3, pandas 2.2.3, scipy 1.18.1, and scikit-learn 1.7.2. Global seed 20260928.

Execution and regression checks.

- The primary benchmark completed 90 primary model fits and 30 XJTU `initial_test` secondary fits, and all hard regression tests passed.
- The model hyperparameter plan and preprocessing plan match the SHA-256 hashes recorded by the primary benchmark regression tests.
- Within each dataset and analysis, reference comparisons used identical eligible physical observations, and no same-cell train/test overlap occurred. All 120 imputer and scaler fits used current training-fold rows only.
- The robustness run completed 270 additional fits under the same model parameters and folds, and all robustness regression tests passed.
- A targeted XJTU A8 reconstruction reproduced all nine stored median cell MAE_Q values to machine precision. It generated the corrected 5,000-replicate Cell_ID bootstrap using one shared resample matrix across reference definitions and verified that the statistic is the difference of model-specific medians, not the median of within-cell paired differences.
- No pipeline replaced a censored endpoint with a point EOL value, generated point RUL for a censored cell, or used target-derived or future-life variables as predictors.

Supplementary Note S2. Endpoint classification rules, classifier provenance, and endpoint audit

S2.1 Estimand

The endpoint estimand is the observed first threshold hit on the retained eligible observation sequence, $N_{\mathrm{obs}}(T) = \min\{k \in \mathcal{K}: Q_k \leq T\}$, with $T = \theta Q_{\mathrm{ref}}$ compared unrounded. The latent

physical first passage over all physical cycles is not reconstructed, and the classifiers contain no monotonicity, persistence, consecutive-below, or rebound-suppression rule.

S2.2 Endpoint classification rules

Classification is a separate step from $N_{\mathrm{obs}}(T)$. The first observed hit is a known quantity in the retained data; the class describes how precisely the declared endpoint grid localizes the crossing around it. Observations of each cell are ordered on the declared endpoint grid: the cumulative aging-operation count for XJTU (ties ordered by raw cycle index) and the raw cycle index for HUST. Let k^- be the retained observation immediately before $N_{\mathrm{obs}}(T)$.

- RIGHT_CENSORED: $N_{\mathrm{obs}}(T)$ does not exist.
- EXACT_OBSERVED: $N_{\mathrm{obs}}(T)$ exists and k^- is adjacent to it on the declared grid.
- INTERVAL_CENSORED: $N_{\mathrm{obs}}(T)$ exists but there is an unobserved index gap between k^- and $N_{\mathrm{obs}}(T)$.

The executable predicates of the threshold-grid classifier, checked in order, are:

Condition (checked in order)	XJTU	HUST
Q_{ref} unavailable	QREF_UNAVAILABLE (no fallback, no imputation)	not applicable (no HUST initial_test)
No retained observation with $Q_k \le T$	RIGHT_CENSORED	RIGHT_CENSORED
First retained observation already $\le T$	LEFT_CENSORED_OR_BASELINE_UNRESOLVED	LEFT_CENSORED_OR_BASELINE_UNRESOLVED
Dense batch (Batch 1 or 2) and index gap from $k^- \le 1$	EXACT_OBSERVED	—
Raw-cycle gap from $k^- \le 1$	—	EXACT_OBSERVED
Otherwise	INTERVAL_CENSORED (bracket k^- to $N_{\mathrm{obs}}(T)$)	INTERVAL_CENSORED

Protocol and QC flags do not decide exactness inside the classifier; they define the upstream retained population. The XJTU dense-batch condition did not change any classification relative to the pure adjacency rule in the frozen outputs: every exact endpoint had a gap of one, all XJTU exact endpoints came from Batches 1–2, and every interval-censored endpoint had a gap of more than one (all in Batches 3–6).

S2.3 Classifier provenance

Component	Executed function	SHA-256 of source file
XJTU Final V2, Batches 2–6 (80% primary)	xjtu_multibatch/analysis.py::detect_eol (dense only for Batch 2)	e428cef82e672b165af5c5a91e466aec90358f1ed959b87e555edae0fc646949
XJTU Final V2, Batch 1 preserved (80% primary)	xjtu_batch1_pipeline/…/sensitivity.py::_detect_eol	db8b46c3e6e331b2d30e20076122c8722784d64dd4b6e245c435199b15f37a75
HUST Final V1 (80% primary)	hust_repro/pilot.py::detect_eol (no gap or baseline branch)	515a7ef5cb7f6c54c9f4e5ee8faaed9199d422f6495910cff9c310bfa2106e8f
75/80/85% grid (Tables 1–2)	detect_xjtu_eol, detect_hust_eol (dense XJTU batches 1 and 2)	880a12834a427b85c3409d91a6503eeb4a0e3f20cfabd90a1821b6bcf03d483f

The manuscript describes a single threshold-grid classifier because all reported endpoint classes come from it. The HUST 80% primary output was produced earlier by the HUST Final V1 function, which has no gap or baseline branch, and the XJTU 80% primary output by the Final V2 functions listed above. At 80% with the common references, the threshold-grid classifier reproduces both primary outputs with zero mismatches (XJTU 0; HUST 0), so the HUST Final V1 classifier and the threshold-grid classifier give identical statuses on the existing samples. No frozen HUST case exercised the gap or baseline branches added in the threshold-grid classifier.

S2.4 Endpoint audit results

A read-only audit traced 1,353 settings (dataset × cell × reference × threshold; XJTU 660, HUST 693). Status-predicate reproduction mismatches: 0. Manuscript-table count mismatches (exact, interval, right, cell count, reference availability, point-RUL eligibility): 0. Left-censored or baseline-unresolved settings: 0. The eight `initial_test` QREF_UNAVAILABLE cells are the XJTU Batch-5 random-walk cells.

Dataset	Status	Settings	Gap (min/med/max)	Later rebound
HUST	EXACT_OBSERVED	503	1 / 1 / 1	5
XJTU	EXACT_OBSERVED	87	1 / 1 / 1	2
XJTU	INTERVAL_CENSORED	170	5 / 6 / 11	3

All 87 XJTU exact settings come from Batches 1–2, and all 170 XJTU interval-censored settings come from Batches 3–6. Gap is the raw-cycle gap from k^- to $N_{\mathrm{obs}}(T)$. Later rebound counts settings in which capacity later returned above the threshold. A later return above the threshold does not undo the observed first hit, but it shows that threshold membership is not absorbing.

Worked examples from frozen outputs.

- EXACT_OBSERVED: HUST_1-1, $\theta = 0.75$, `first_comparable`, threshold 0.8807244 Ah. The preceding retained observation is cycle 1502 (0.8810301 Ah) and the first hit is cycle 1503 (0.8804143 Ah), a raw-cycle gap of one.
- INTERVAL_CENSORED: XJTU B3_R2.5_battery-1, $\theta = 0.80$, `nominal`, threshold 1.6 Ah. The preceding retained observation is at cumulative aging-operation count 572 (raw cycle 573; 1.607 Ah), and the first hit is at cumulative aging-operation count 577 (raw cycle 578; 1.599 Ah), in a sparse-diagnostic batch.
- RIGHT_CENSORED: HUST_1-1, $\theta = 0.75$, `nominal`, threshold 0.825 Ah. The last retained observation is cycle 1504 (0.8804153 Ah), above the threshold.

S2.5 Historical per-cycle RUL exports

The manuscript reports only cell-level exact point-RUL eligibility. Earlier per-cycle RUL exports used different post-endpoint policies, which are documented here for transparency and do not affect any manuscript count.

Export	Rule	Post-endpoint rows
XJTU Batch 1 (frozen)	RUL emitted only for Aging_Cycle_Index ≤ EOL aging cycle	44 rows marked POST_EOL_EXCLUDED with blank RUL
XJTU Final V2, Batches 2–6	RUL = max(0, exact EOL aging cycle − current aging cycle)	7 Batch-2 rows clipped to zero
HUST Final V1	RUL = EOL cycle − raw cycle for Raw_Cycle_Index ≤ EOL cycle	none (rows after the endpoint omitted)

The XJTU Final V2 export has 41,708 rows with 7,956 numeric labels, and the HUST Final V1 export has 550,925 rows with 258,681 numeric labels. A single post-endpoint policy would need to be approved before a manuscript-wide per-cycle RUL label count is reported.